Advanced Model Risk Management

Tutorials 1–2 — The Phoenix Autocall as a Case Study

Two 90-minute sessions — Notebook: `TD_model_risk.ipynb` (posted on the ENT)

Foreword: why this product

Model risk is hard to teach because most examples are degenerate: either the model is obviously wrong, or the product is so simple that every model agrees. The Phoenix autocall avoids both traps. It is one of the few instruments for which *several models calibrate to exactly the same market data and still give different prices.* The divergence therefore comes neither from an implementation error nor from a bad set of parameters: it is structural, and that is precisely the definition of model risk.

The tutorial follows seven steps. We start from the **payoff** and its decomposition, infer **which risks** the product carries, then **which model** is acceptable, and then turn the question around by judging models no longer on their prices but on **how faithful their sensitivities are**. We end with two warnings: faithful greeks are not a sufficient criterion, and enriching a model is not always the right answer.

How the tutorials work

Vocabulary. TD = tutorial session: TD 1 and TD 2 are the two 90-minute sessions.
Teams. You work in **teams of 4 to 5**. Each team has its own **term sheet**: a coupon barrier B_c and a protection barrier B_p, given by your group and team number in the team table (Excel answer sheet, tab *Teams*, and notebook §0). Every number you report is for your own term sheet.
The parts are independent. Sections 1 to 8 and Block D can be read and answered in any order: each question recalls the result it needs, and the numbers it relies on come from the notebook. The only prerequisite is to run notebook §0 with your group and team number.
Who answers what. Each question carries a symbol:

•	**Team**	Answered by the team in the team answer sheet (Excel), including its *Decision* tab.
♦	**Individual, mandatory**	Answered by *each student alone*, in the individual answer sheet (Excel).
◦	**Optional**	For those who want to go further; not graded.

Before TD 1: read the subject. Please take the time to read this subject before the first session: at least Section 1 (the product, the definitions, the parameters and the reminders (R1)–(R9)), and skim the questions. Also run the notebook once at home (∼3 minutes) to check that it works. You will lose much less time in class.

Phase	Questions	What happens
Before	Read the subject (at least Section 1) + run the notebook once	Not graded. Check at home that the notebook runs.
TD 1 (90 min)	Q10, 11, 12, 13, 16, 14	The model gap, on your term sheet. Fill in part 1 of the answer sheet (kept by the team).
TD 2 (90 min)	Q18, 22–26 + Block D (D1–D3) + *Decision* tab	What to do with the gap; a pricer written by an AI; your validation decision. Teams that have finished may start the individual part on their own.
After	Team: finish the answer sheet (one week). Individual: all ♦ questions (end of October)	Team answer sheet and notebook; individual answer sheet (+ optional PDF appendix).

Roadmap of the two sessions (minutes)

TD 1 (whole team). 0–5: set `GROUP` and `TEAM` in §0, run the *whole* notebook (∼3 min), and meanwhile read Section 1.3 and, in Section 1.4, (R1), (R4) and (R6); designate your **Block D lead** and enter his or her email in the answer sheet (tab *Team*). **5–15**: Q10. **15–28**: Q11. **28–52**: Q12–13. **52–72**: Q16. **72–82**: Q14 (shorten it if you are running late). **82–90**: answer sheet, part 1 (Excel template). Nothing is submitted at the end of TD 1.

TD 2 (the team splits in two). 0–3: recap of model M4 by the instructor. *Block D sub-team (the lead and one or two members).* **3–13**: test plan (D1), handed in *before* receiving the module; the instructor then emails your team's module to the Block D lead, who shares it with the team. **13–50**: run the tests (D2) and measure the impact of each error (D3). *Rest of the team.* **3–15**: Q18. **15–30**: Q22, Q23, Q24. **30–50**: Q25–26 (notebook §7, run by you). *Whole team.* **50–75**: the *Decision* tab (Section 10). **75–90**: complete the answer sheet; teams that have finished start the individual part on their own. Nothing is submitted at the end of a session: see the deadlines below.

Precision. Every number you report comes with the number of paths and a standard error (printed by the notebook for prices and break-even coupons), or failing that a second run with another seed.

Setup. Python ≥ 3.9 with numpy, scipy and matplotlib (recent numpy: see the first cell of the notebook).

Grading: 50 % team, 50 % individual. There is no oral presentation. *Team work* (50 %): the team answer sheet `TD_answer_sheet.xlsx` — numbers, precision and Block D tests (25 %); *Decision* tab (25 %). *Individual work* (50 %): answers to the ♦ questions in the individual answer sheet `TD_individual_answers.xlsx`, written alone; long derivations may go in an optional PDF appendix. Everything is submitted on **Moodle** (or by email to the instructor if Moodle is unavailable). Groups IF3 and IF2, whose two tutorials run back to back on the same evening, are graded with the same leniency for this atypical schedule. The detailed grid is in the **evaluation sheet**.

Group	**Team answer sheet + notebook** (one week after TD 2)	**Individual answer sheet (+ PDF appendix)** (end of the month)
IF3	Fri, Oct 9, 11:59 pm	Sat, Oct 31, 11:59 pm
IF2	Thu, Oct 22, 11:59 pm	Sat, Oct 31, 11:59 pm
IF1	Fri, Oct 23, 11:59 pm	Sat, Oct 31, 11:59 pm

Several questions explicitly recall the result to use: they should not block anyone. The technical terms of Section 1.2 (EMTN, *worst-of*, *funding*...) are not needed in class.

1 The product and what is at stake

1.1 Mechanics

A *Phoenix autocall* is a debt security (typically an EMTN) whose cash flows are linked to an equity underlying and observed on fixed dates. Three building blocks, checked on each date t_i:

- **Conditional coupon**: paid if $S_{t_i} \geq B_c S_0$ (coupon barrier, e.g. 70 %), with or without a *memory* feature (catch-up of missed coupons).
- **Early redemption** (*autocall*): if $S_{t_i} \geq B_a S_0$ (e.g. 100 %), the note is redeemed at par and all future cash flows disappear.
- **Conditional protection**: at maturity, redemption at par unless $S_T < B_p S_0$ (e.g. 60 %), in which case the investor bears the underlying's performance.

Definitions used throughout

At par. At 100 % of the notional. The note is issued at par (the investor pays 100) and, when it is called or protected, redeemed at par (the investor receives 100). *Below par* or *above par* means a price under or over 100 %.
Expected life (expected duration). The redemption date τ of an autocall is random: it depends on the path of the underlying. Its *expected life* is $\mathbb{E}^{\mathbb{Q}}[\tau]$, the average time the note stays alive. A plain bond has a known life; an autocall's life shortens when early redemption becomes more likely.
Separable instrument. A feature of a contract is *separable* when it could be detached and traded on its own, as a stand-alone contract with its own cash flows, without changing the rest of the contract (the accounting notion of an embedded derivative that can be split from its host).
Structuring equation. The identity that, at issue, splits the issue price (par) among the building blocks of the product and the fees kept by the bank: it shows what finances the coupon.
Observed at each close. A barrier checked against the official *closing price* of the underlying on every trading day, rather than only at maturity (European barrier) or at every instant during the day (continuous barrier).

1.2 Economics

What a Phoenix really is

The coupon is not a yield: it is the price of an option sold by the investor. The product is a **bet that the market will not fall**, never a bet that it will rise — there is no participation in the upside. The distribution of outcomes is highly skewed: many small gains, few losses but very heavy ones. The median is flattering; the mean much less so.

Five points shape the economics of the product and will come back in the tutorial.

1. **Where the budget comes from.** The coupon is financed by the premium of the put sold by the investor and by the issuer's *funding* benefit (the zero-coupon bond is issued on the issuer's credit curve). Rate levels and credit spreads therefore drive how generous the coupon can be as much as the level of volatility does.
2. **Early redemption shortens the expected life.** The most likely scenario (a stable or rising market) is also the one that pays least in absolute terms: an early call, one or two coupons, reinvestment risk.

3. **The choice of underlying dominates everything else.** A *worst-of* on several uncorrelated stocks, or a *decrement* index (a fixed synthetic dividend subtracted, hence a lower forward), raise the displayed coupon far more than any barrier setting. They raise the risk in the same proportion.

4. **The issuer's credit risk is real.** "Capital protection" is only as good as the issuer's signature.

5. **The secondary market is a monopoly.** The issuer is usually the only market maker, and distribution fees, paid upfront, are amortized in the exit price.

1.3 Parameters used in the tutorial

Underlying, initial spot S_0	single stock, 100
Maturity T	3 years
Observation dates	semi-annual: 0.5, 1.0, …, 3.0
Autocall barrier B_a	100 %
Coupon barrier B_c	70 % (reference case; your team: 65, 70 or 75 %)
Protection barrier B_p	60 % (reference case; your team: 50, 55 or 65 %)
Memory feature	no (unless stated otherwise)
Risk-free rate r, dividend yield q	2 %, 2 %
Issue price	at par (100 %)

Your term sheet. Wherever the subject says 70 % (coupon) or 60 % (protection), read the barriers of your team's term sheet; in the notebook, set `GROUP` and `TEAM` in the parameter cell of §0 and the barriers follow. The numbers quoted in this subject are those of the reference case.

The **market** is represented by a reference Heston model ($v_0 = \theta = 4\,\%$, $\kappa = 1.5$, $\xi = 0.6$, $\rho = -0.7$), from which the whole implied volatility surface is derived. The 3-year ATM implied volatility is about 17 %, with a pronounced skew.

1.4 Reminders, notation and results taken as given

This section gathers everything used in the tutorial. **None of these results needs to be proved again**: they are provided to be used.

(R1) Risk-neutral price and stopping time. Under a risk-neutral measure $\mathbb{Q}$, writing $R_i = S_{t_i}/S_0$ and

$$\tau \;=\; \inf\{\, t_i,\; i < n \;:\; R_i \geq B_a \,\}, \qquad \inf \emptyset := t_n,$$

the price of the Phoenix reads

$$V_0 \;=\; \mathbb{E}^{\mathbb{Q}}\left[\sum_{i=1}^{n} e^{-rt_i}\, c\, \mathbf{1}_{\{R_i \geq B_c\}}\, \mathbf{1}_{\{t_i \leq \tau\}} \;+\; e^{-r\tau}\, \Pi_\tau \right], \quad \Pi_\tau = \begin{cases} 1 & \tau < t_n \\ 1 - (1 - R_n)\, \mathbf{1}_{\{R_n < B_p\}} & \tau = t_n. \end{cases}$$

(R2) The four models. All under $\mathbb{Q}$, with S_0 and r, q given.

$$\textbf{M1}: \ \frac{dS_t}{S_t} = (r-q)\,dt + \sigma_{\text{ATM}}\,dW_t \qquad \textbf{M2}: \ \frac{dS_t}{S_t} = (r-q)\,dt + \sigma_{\text{loc}}(S_t,t)\,dW_t$$

$$\textbf{M3}: \ \frac{dS_t}{S_t} = (r-q)\,dt + \sqrt{v_t}\,dW_t^1, \quad dv_t = \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^2, \quad d\langle W^1, W^2\rangle_t = \rho\,dt.$$

$$\textbf{M4 (local-stochastic)}: \ \frac{dS_t}{S_t} = (r-q)\,dt + L(S_t,t)\sqrt{v_t}\,dW_t^1,$$
$$dv_t = \kappa(\theta - v_t)\,dt + \boldsymbol{\eta}\,\xi\sqrt{v_t}\,dW_t^2,$$

where L is the **leverage function** and $\eta \in [0,1]$ the **mixing fraction**. M4 is the market standard for autocalls; it was only mentioned in the lectures. *What to remember:* at each date, L is chosen so that $L^2(S,t)\,\mathbb{E}^{\mathbb{Q}}[v_t \mid S_t = S] = \sigma_{\text{loc}}^2(S,t)$; by Gyöngy (R4), every value of η therefore reproduces *all* vanillas. η sets the share of stochastic volatility: $\eta = 0$ gives back local volatility M2, $\eta = 1$ stochastic volatility M3. Vanillas do not see η; the forward skew does. The notebook provides M4 already calibrated.

(R3) Dupire's formula. In total implied variance $w(y,T) = \sigma_{\text{imp}}^2(y,T)\,T$ with $y = \ln(K/F_T)$:

$$\sigma_{\text{loc}}^2(y,T) \;=\; \frac{\partial_T w}{1 - \frac{y}{w}\,\partial_y w + \frac{1}{4}\left(-\frac{1}{4} - \frac{1}{w} + \frac{y^2}{w^2}\right)(\partial_y w)^2 + \frac{1}{2}\,\partial_{yy} w}.$$

(R4) Gyöngy's theorem (1986), taken as given. Let S be a stochastic volatility diffusion. Setting $\sigma_{\text{loc}}^2(K,T) = \mathbb{E}^{\mathbb{Q}}[v_T \mid S_T = K]$, the process $\widetilde{S}$ solving $d\widetilde{S}_t/\widetilde{S}_t = (r-q)\,dt + \sigma_{\text{loc}}(\widetilde{S}_t,t)\,dW_t$ has, *for each* T, the same marginal law as S_T. Nothing is claimed about the joint laws of $(S_{T_1}, \ldots, S_{T_k})$.

(R5) Completeness criterion, taken as given. A market driven by d independent Brownian motions is complete if d traded assets (besides the numeraire) are available whose volatility matrix is invertible. In this tutorial, unless stated otherwise, we hedge with the *underlying alone* (and cash). Otherwise, the martingale measure is not unique and some payoffs cannot be replicated.

(R6) Forward variance and forward smile. The forward variance is $\mathbb{E}^{\mathbb{Q}}\big[\int_{T_1}^{T_2} v_s\,ds \,\big|\, \mathcal{F}_0\big]$. The **forward smile** $T_1 \to T_2$ is the implied volatility surface of *forward-starting* options, with payoff $(S_{T_2}/S_{T_1} - k)^+$. Its slope is the **forward skew**. It depends on the law of S_{T_2} *given* S_{T_1} (the *transition kernel*), which calibration to vanillas does not constrain.

(R7) The three valuation layers, and what they correct.

Layer	Framework	What it corrects	Where it is booked
Front-office valuation	internal (management)	nothing: it is the desk model's *mid* price	management P&L, greeks
Fair value adjustments	**accounting**: IFRS 13, IFRS 9	the gap between the *mid* price and the *exit price* of an actual position	income statement and balance sheet (fair value)
Prudent valuation (AVA)	**prudential**: CRR art. 105, EBA RTS	the gap between fair value and a *prudent* value (90 % confidence)	deduction from CET1 capital

IPV is not a reserve. *Independent Price Verification* (CRR art. 105(8)) is a **control**: a periodic check — at least monthly — of the accuracy of the front office's parameters and marks against independent sources. It produces *differences*, which may trigger an adjustment; it is not one.

Usual fair value adjustments (accounting): bid-offer / close-out cost; parameter uncertainty (unobservable inputs); model uncertainty; concentration and liquidity; XVA (CVA, DVA, FVA); and, for Level 3 positions, **deferral of day-one P&L** (IFRS 9) — which is not a reserve but an accounting rule.

AVA categories (prudential): market price uncertainty — close-out costs — **model risk** — concentrated positions — future administrative costs — early termination — operational risk — investing and funding costs — unearned credit spreads. AVAs **come on top of** the accounting adjustments already booked: they are added to them, and a risk already covered by an accounting adjustment is not counted a second time.

Three kinds of uncertainty not to be confused: *(i)* a poorly observed **market** parameter — the quantity exists and trades, but without a reliable consensus (long-dated dividends, repo, implied correlation of a basket); *(ii)* a **model** parameter — the quantity exists only inside the model and no data can reveal it (η); *(iii)* **specification** risk — the choice of the model family itself. The three call for different treatments and governance.

(R8) Vocabulary. *Vega convexity* (or *volga*): $\partial^2 V/\partial\sigma^2$, zero for an at-the-money vanilla, nonzero elsewhere. — *Vega buckets*: breakdown of vega by maturity bucket. — *Day-one P&L*: profit recognized at inception, equal to the gap between the transaction price and the model value. — *AVA* (*Additional Valuation Adjustment*): see (R7). — *Cross gamma*: second-order cross sensitivity $\partial^2 V/\partial x\,\partial y$ between two risk factors. — *Gap risk*: loss caused by a move of the underlying across a level where the payoff is discontinuous (a barrier), between two rebalancings or through a jump. Near the barrier and maturity, the delta becomes so large that the hedge cannot follow: the loss is of the order of the payoff's jump, whatever the model. — *Delta* (notebook, §7): change in value for a 100 % rise of the spot, as a fraction of the notional, i.e. $S\,\partial V/\partial S$; a delta of 20 means that a 1 % move of the spot changes the value by 20 % of the notional.

(R9) What a model is (supervisory definition). A *model* is a quantitative method, system or approach that applies statistical, economic, financial or mathematical theories, techniques and assumptions to process input data into quantitative estimates. It has three components: an input component, a processing component that turns inputs into estimates, and a reporting component. Who or what wrote the code plays no role in this definition. Model changes are graded by materiality, as in common bank policies (Session 5): **I — minor** (routine recalibration, data refresh, cosmetic code change): notification and automated tests; **II — material** (new product in scope, new calibration set, library upgrade): targeted review before use; **III — major** (new methodology, new model, new numerical scheme): full validation before use.

Convention: the break-even coupon

Coupons are paid **semi-annually**. The **break-even coupon** is the semi-annual coupon c such that the value of the product equals par. Since the value is affine in c,

$$c \;=\; \frac{1-(\text{redemption leg})}{\text{annuity}}, \qquad \text{annuity} = \sum_i \mathbb{E}^{\mathbb{Q}}\big[e^{-rt_i}\,\mathbf{1}_{\{\text{coupon paid at } t_i\}}\big].$$

It is quoted **per year**: the annual coupon is $2c$, as on a term sheet ("6 % p.a., paid semi-annually"). Coupons are paid out, not reinvested, so there is no compounding: not $(1+c)^2-1$. Every model gap is expressed in **basis points of annual coupon** (twice the semi-annual gap), the one unit that speaks to the distributor and to the trader alike.

2 Decomposing the payoff

Q1. ♦ Write the cash flow of the product on date t_i as a function of S_{t_i} and of the survival indicator $\mathbf{1}_{\{\text{not called before } t_i\}}$. Deduce a decomposition of the Phoenix into four elementary building blocks: a zero-coupon bond, a sold barrier put, a strip of bought digitals, and a *knock-out* condition.

Q2. ♦ **Show that** the price of the product reads as in (R1), explaining the role of each indicator. **Deduce that** the payoff is not a function of S_T alone: state the variables it depends on. *(The result to establish is given in (R1); the task is to justify it term by term.)*

Q3. ♦ Explain why the autocall clause is not a separable instrument (see the definitions in Section 1.1) but a condition on *all* future cash flows. What does this imply for the expected life of the product?

Q4. ♦ Write the structuring equation linking the coupon, the put premium, the cost of the digital strip, the funding benefit, the structuring margin and the distribution fee.

Q5. ♦ The put sold here has a **European barrier**: the protection barrier is only observed at maturity. Write its payoff. How does it differ from a *down-and-in* put with an American barrier (observed continuously, or at each close)? Which of the two is more expensive, and why did the European market switch massively to the first one after 2008?

3 A taxonomy of model risk and a decision ladder

We use the usual three-way taxonomy: **(i)** the model is badly implemented; **(ii)** the model is correct but fed with bad inputs; **(iii)** the chosen model is not the right one for this derivative. Form (i) is the subject of Block D (Section 9); everywhere else, implementations are assumed to be sound.

Q6. ♦ For each of the three forms, give a concrete example drawn from the Phoenix.

Q7. ♦ Consider the following **decision ladder**, which ranks payoffs by increasing demands on the model:

Level	Nature of the payoff	Minimal model
A	decomposable into vanillas	Black–Scholes, one implied vol per strike
B	sensitive to the skew, not decomposable	properly calibrated local vol
C	vega convexity, forward skew	stochastic volatility

Rank: (a) a call spread; (b) an Asian option; (c) a European digital; (d) a European-barrier put; (e) a down-and-in put with a continuous barrier; (f) a cliquet with local floors and caps; (g) the full Phoenix. Justify each answer in one sentence.

Q8. ♦ Why can a call spread be priced in Black–Scholes although it is sensitive to the skew? What in the Phoenix prevents the same reasoning?

Q9. ♦ **Using** the dynamics (R2) and the completeness criterion (R5): for each of the three models, give the number of Brownian motions and the number of traded assets when hedging with the *underlying alone*. **Deduce that** M1 and M2 then define complete markets, and M3 an incomplete one. What does this imply in practice for what one can expect from a delta hedge under M3? What would adding vanillas to the hedging instruments change?

4 Calibration *is* a model choice

This is the heart of the tutorial. We have three models: **(M1)** Black–Scholes with a flat volatility set at the 3-year ATM; **(M2)** Dupire local volatility calibrated to the whole surface; **(M3)** the reference stochastic volatility model. By construction, M2 and M3 reproduce *exactly the same* vanilla prices.

Q10. • **Using Gyöngy's theorem (R4)**, show that M2 and M3 give the *same* value to any payoff of the form $g(S_T)$. Check this result numerically **(notebook, §1–2)** on call prices. Finally, **show that** the argument does not extend to a payoff of the form $g(S_{T_1}, S_{T_2})$: which object, not constrained by (R4), then comes in? *(Hint: see (R6).)*

Q11. • **(Notebook, §3)** Price under M1, M2 and M3 the *European-barrier put* alone (strike 100 %, barrier B_p of your term sheet, maturity 3 years). Comment on the gap between M2 and M3 on the one hand, and on the gap with M1 on the other. What do you conclude about the role of the model choice for a payoff whose value depends only on S_T?

Q12. • **(Notebook, §4)** Now price the **full Phoenix** under the three models and compute the break-even coupon of each. Express the M2/M3 gap in basis points per year.

Q13. • Where does this gap come from, given that it was essentially zero in Q11 (the put alone)? Identify precisely the feature of the payoff responsible, then the **two channels** through which it acts: (a) what the product becomes after a date on which it is not called; (b) the probability of being called on each date, hence the expected life of the product (compare the annuities printed by the notebook).

Q14. • **(Notebook, §4)** Redo the computation with a protection barrier *observed at each close* (see Section 1.1), and no longer at maturity only. By how much does the break-even coupon change? Does the M2–M3 gap widen? What calibration does this product require: exact at maturity only, or smooth over the whole surface (strikes and maturities)? *(Hint: how is the vega spread over time?)* Finally, what must the term sheet specify?

Q15. ♦ A structurer prices the product in local volatility and quotes the corresponding coupon. The structurer wins the deal against a competitor who prices in stochastic volatility. Describe the structurer's economic position. Is it a gain?

Q16. • **(Notebook, §4bis)** We measure the *forward smile* 1 year → 3 years in the sense of (R6), under M2 and under M3, together with the spot smile of the same length (from 0 to 2 years). The *slope* is the implied vol at strike 70 % minus the implied vol at strike 110 %, in vol points. Compare the three slopes, and say why they are the same for every

term sheet. **Deduce** the precise mechanism behind the gap found in Q12, and explain why calibration to vanillas does not detect it. *(Hint: after a date on which the product is not called, what does the remaining product look like?)*

Q17. ∘ Model M4 (R2) is the one desks actually use. **Using Gyöngy's theorem (R4)**, show that exact calibration to vanillas requires

$$L^2(S,t)\,\mathbb{E}^{\mathbb{Q}}[v_t \mid S_t = S] \;=\; \sigma^2_{\mathrm{loc}}(S,t).$$

Check that $\eta = 0$ gives back M2 and $\eta = 1$ gives back M3. Why does this condition require calibrating L and the simulation *simultaneously* (the so-called particle method)?

Q18. • **(Notebook, §4ter) Validate M4 before using it.** M4 is provided calibrated. For $\eta \in \{0; 0.25; 0.5; 0.75; 1\}$: (a) check that the vanillas are reproduced in every case; (b) **check the bounds**: does $\eta = 0$ give back the coupon of M2, and $\eta = 1$ that of M3? Quantify the gap at each bound and say whether it comes from the model or from the code; (c) record the break-even coupon and the forward smile slope, then give the **coupon range** as η spans $[0, 1]$, with its uncertainty. **What do you conclude** about whether η can be identified from market data? *The operational definition of the model uncertainty adjustment that follows, and why it is more satisfactory than the M2/M3 gap of Q12, go in the* Decision *tab (Q36).*

5 The sign of vega and skew: the trap

Q19. ♦ The seller of the Phoenix is short the digital strip and long the barrier put. Reasoning on where the forward stands relative to the coupon barrier, determine the sign of the volatility sensitivity of *each* of the two components.

Q20. ♦ **(Notebook, §5)** Compute numerically the vega of each building block, then of the whole product. Does the result confirm the intuition? What happens to the sign if the coupon barrier is raised from 70 % to 95 %?

Q21. ♦ Same question for the skew. What *pricing* practice does a prudent trader adopt when the net sign of a parameter is not stable?

6 Choosing a model by the faithfulness of its sensitivities

So far, models have been judged on their *prices*. We now turn the question around: a model is not a price generator, it is a generator of **hedging strategies**. The right model is the one whose greeks allow the payoff to be replicated.

Q22. • **(Notebook, §6)** Assume that "reality" follows the reference model M3. Delta-hedge the European-barrier put, rebalancing every week for 3 years, with the deltas of M1, M2 and M3 in turn. Compare the mean, the standard deviation and the 5 % quantile of the terminal hedging P&L.

Q23. • Interpret the *mean* and the *standard deviation* of the P&L separately. Which one measures a price error, which one a greeks error? *(Discussed in class, orally.)*

Q24. • M2 and M3 show the same initial price. Does their hedging performance tell them apart? What dominates the residual risk? Two leads: (a) the M3 delta in the notebook is computed at fixed variance v; when $\rho \neq 0$, is it the delta that minimizes the

variance of the P&L (see the market regime and the SSR, Session 4)? (b) would this residual risk disappear with a better model, with other hedging instruments (vanillas), or with neither? Deduce whether it is a "model" risk or whether it comes from market incompleteness when hedging with the underlying alone.

Q25. • **(Notebook, §7, run by your team)** Plot the delta of the Phoenix around the protection barrier B_p at $t = 2.5;\ 2.9;\ 2.99;\ 2.999$ (unit: see (R8)). Comment. Can a desk execute this hedge? This is the *gap risk* of (R8). *Validator's remark:* the value of the delta very close to maturity depends on the computational grid; why should you distrust it, and which value would you retain?

Q26. • **(Notebook, §7)** Replace each barrier indicator with a call spread of half-width ε (*barrier shift*). Record the maximum delta as a function of ε (notebook table). Then compute the effect of ε on the break-even coupon. What do you observe, and why does *symmetric* smoothing generate almost no margin? How does a desk actually proceed? *(Hint: on which side should the barrier be shifted to be prudent?)*

Q27. ○ Also plot the delta around the autocall barrier just before an observation date. What happens and how should it be interpreted economically?

Caveat to state explicitly

Faithful greeks are a **necessary but not sufficient** criterion, for at least three reasons: (a) a "reality" has to be assumed to assess them, and that is exactly the unknown; (b) part of the residual risk cannot be reduced by a better model: the volatility left unhedged when hedging with the underlying alone (Q24), and the gap risk near the barrier (Q25); (c) a model can hedge well and price badly, or the reverse. The hedging criterion therefore does not replace the pricing criterion: it complements it.

7 Front office, independent control, prudent value

The same product is valued at several levels, by functions with different goals and under different frameworks. The point not to miss: the **accounting** and the **prudential** layers do not correct the same thing and are not booked in the same place (R7).

Q28. • Complete and comment on the following table.

	Front office	**Control (IPV)**	**Prudent value**
Main objective	?	?	?
Framework	?	?	?
Success criterion	?	?	?
Frequency	?	?	?
Treatment of barriers	?	?	?
Attitude toward complexity	?	?	?

Q29. • **Using (R7)**, explain why IPV is *not* a reserve, and what it produces instead. What happens when an IPV difference is found and judged significant?

Q30. • Distinguish precisely a **fair value adjustment** from an **AVA**: (a) under which framework is each one required; (b) what does each one try to approximate — exit price? prudent value, and at what confidence level?; (c) where is each one booked; (d) AVAs

are added on top of the accounting adjustments already booked: how does one make sure that the same risk is not counted twice?

Q31. • **Classify** each of the uncertainties met in this tutorial according to the three kinds of (R7) — poorly observed market parameter, model parameter, specification — then state the adjustment that most naturally deals with it.

Uncertainty	Kind	Accounting adjustment	AVA category
Mixing fraction η (Q18)	?	?	?
Choice local vol / local-stochastic	?	?	?
Projected dividends at 3 years (single stock)	?	?	?
Implied correlation of a *custom* basket	?	?	?
Volatility left unhedged by the delta alone (Q24)	?	?	?
Gap risk at the barrier (Q25)	?	?	?
Gap between theoretical and smoothed price (Q26)	?	?	?

Q32. • The front office smooths its barriers (*barrier shift*) and hedges the regularized product. Should independent control validate this value? Argue both ways, then say which layer deals with the remaining gap.

Q33. • The front office prices in local volatility; the independent valuation team uses M4. The gap is persistent. **How does it show**, first in the accounting result and then in capital? Why is it not the same amount?

Q34. • A product is sold at 100 while the validated model values it at 98. What happens to the gap of 2 in the accounts if the product is classified as Level 3 in the fair value hierarchy? How does this rule protect against the situation described in Q15?

Q35. • State the central tension in one sentence, contrasting *accuracy of the representation* and *operability of the hedge*. Can it be settled?

Q36. • A desk proposes to replace its local volatility model with model M4, for which Q18 measured the coupon range as the mixing fraction η spans $[0, 1]$ and assessed whether η can be identified from vanillas. Should the validation committee approve? Which safeguards should it require, on what principle should η be set, and to which layer of (R7) should this range be attached?

8 Counter-example: when *not* to enrich the model

Reminder: what the product is, economically

Before tackling this part, keep in mind that the Phoenix is not only a payoff. It is a **debt security** issued on the issuer's funding curve, **redeemed at a random date** that depends on the path of the underlying, sold at a fixed coupon during several weeks of marketing, and then quoted by a single market maker. Its life is not a parameter: it is a random variable. This observation alone opens a dimension of risk that everything above has ignored.

Q37. ♦ The seller hedges the rate leg with zero-coupon bonds. To isolate the mechanism, take the simplest autocall: **maturity 2 years, with a single early-redemption date at**

1 year (the note is called at 1 year if $S_1 \geq B_a S_0$, and otherwise runs to its maturity at 2 years, where it is redeemed). Until the 1-year date, the issuer does not know whether it will repay after 1 year or after 2 years: its rate hedge is a mix of 1-year and 2-year zero-coupon bonds, weighted by the probability of being called. Describe how this rate hedge must be rebalanced in the following two cases: (a) the underlying rises; (b) the underlying falls. Then say what happens to the hedge *at* the 1-year date if the note is called, and *between 1 and 2 years* if it is not.

Q38. ♦ Assuming a *positive* correlation between the underlying and interest rates, show that the rebalancing loses money in both cases. Redo the reasoning with a negative correlation. Deduce the direction of the effect on the price.

Q39. ♦ How does this cost vary with the maturity of the product? With the number of early-redemption dates?

Q40. ♦ Should one therefore adopt a stochastic interest rate model? Build the argument *against*, carefully distinguishing what a model lets you *see* from what it lets you *do*.

Q41. ○ What, then, is market practice? *(Practitioner's order of magnitude, given for guidance only: about 10 bp of coupon per year, depending on the maturity and the size of the deal.)*

Q42. • **Synthesis.** State the general criterion that emerges from the whole tutorial: under what condition is enriching a model justified? Apply this criterion to the four decisions met along the way: (a) moving from Black–Scholes to local volatility; (b) moving from local volatility to stochastic volatility; (c) moving to local-stochastic M4; (d) moving to stochastic rates. *In the Decision tab, only (a) to (c) are graded; (d) draws on the individual questions Q37–Q40.*

9 Block D — A pricer written by an AI: validate it

Why this block

A code assistant produces in a few seconds a clean, documented pricer that runs and prints a plausible coupon. The cost of *production* has collapsed; the cost of *verification* has not moved. The job shifts toward designing tests: this is exactly form (i) of the taxonomy of Section 3, which the rest of the tutorial sets aside.

Organization. At the start of TD 1, each team designates a **Block D lead** and enters his or her email address in the answer sheet (tab *Team*). Once the team has handed in its test plan (D1), the instructor emails the team's module, `phoenix_pricer_llm.py`, to the Block D lead, who shares it with the rest of the team. The module was generated by an LLM from the description of *your* term sheet and of the reference market. It contains several errors. You are the validator. Modules differ from one team to another and from one tutorial group to another.

Interface of the module (all you need to write the plan)

`simulate_heston(S0, T, n_steps, n_paths, r, q, hp, seed=None)`: spot paths, an array of shape $(n_{\text{steps}} + 1) \times n_{\text{paths}}$ (one column per path).
`phoenix_pv(paths, T, coupon, product, r)`: discounted value of the cash flows, path by path (notional 1); `paths` has the same shape as above, and hand-built paths can be passed in.
`price(coupon, S0=None, n_steps=300, n_paths=50_000, seed=None, ...)` (pass arguments by name: the second positional argument is the spot), `break_even_coupon(...)`, `delta(coupon, bump=0.001, ...)`. Constants `PRODUCT`, `MARKET`, `HESTON`. The coupon is *semi-*

annual. The module calls the vol of vol `xi` (the notebook calls it `sigma`): do not pass the notebook's parameters to the module directly.

D1. • **Before receiving the module**, write a test plan (half a page), handed in on Moodle or on paper: **at least five tests, one per error class** (numerical scheme, market data, payoff logic, term-sheet reading, greeks), each with its expected result, justified without the module. Hint: a calibrated model must reproduce a martingale, known vanilla prices, and hand-built paths; a greek must be stable from one computation to the next.

D2. • Run your tests against the module. List those that fail, and locate the line responsible. One error can hide another: once you have found one, fix it and rerun your tests.

D3. • For each error found: fix it, then measure its impact in basis points of annual break-even coupon, *all else being equal* (same random draws).

D4. • The assistant claims that "the break-even coupon it returns is consistent with the literature." What has it actually checked? What should have been checked?

D5. • Is this pricer a *model* in the sense of definition (R9), even though it was written by a generative AI tool? Who must validate it, under which change category (I to III, see (R9)), and how can one prevent such a tool from circulating outside the model inventory (*shadow IT*)? *(Everything you need is in (R9); no external text is required.)*

Rule of the game

Block D is the only part where an AI may be used, and only to help write your tests. You remain solely responsible for what you hand in: the *Decision* tab states which tools were used for the tests and what you checked yourselves. Everywhere else, AI tools are not allowed.

10 The *Decision* tab (TD 2, team)

There is no oral presentation: the team's validation decision is written in the *Decision* tab of the team answer sheet. It contains, in short answers:

- **the decision**, for the model *and* for the Block D pricer: approve, approve with conditions, or reject, justified in five lines with the numbers of your term sheet and their precision, and what would change it;
- **the model uncertainty adjustment**: the η range retained (Q18) and the principle used to set η (Q36);
- **Section 7**: Q28 to Q35;
- **Block D**: D4 and D5 (the impacts of D3 are in the *Block D* tab);
- **the criterion** of Q42, parts (a) to (c);
- **Block D AI statement**: the AI tools used to write your Block D tests, if any, and what you checked yourselves (or "none").

Start it in TD 2 (minutes 50–75); the team has one week after TD 2 to finish and submit it.

Appendix — The two answer sheets

The answer sheet is an **Excel file**, `TD_answer_sheet.xlsx`, posted on the ENT. One row per number, for *your* term sheet; add rows if needed. It is filled in during both tutorials and submitted on Moodle one week after TD 2, together with the notebook. Its tabs: *Team* (group, team number, members, Block D lead's email), *Answers* (question, numbers, paths, standard error, one-sentence reading), *Block D* (test plan and results), *Decision* (Section 10), *Teams* (term sheet of every team).

The **individual answer sheet**, `TD_individual_answers.xlsx`, has one cell per ♦ question. Write formulas in plain text (e.g. `V0 = E[sum exp(-r t_i) c 1{R_i >= Bc} ...]`); long derivations may go in an optional PDF appendix, submitted with it.

Useful references: M. Bouzoubaa & A. Osseiran, *Exotic Options and Hybrids*, Wiley, 2010 (ch. 4, 10, 11, 12); J. Gatheral, *The Volatility Surface*, Wiley, 2006; L. Bergomi, *Stochastic Volatility Modeling*, CRC, 2016; Basel Committee / CRR art. 105 on *prudent valuation*; J. Guyon & P. Henry-Labordère, *Nonlinear Option Pricing*, CRC, 2014 (local-stochastic model, particle method).